

1 NC-smooth affine schemes (following Kapranov)

We start by recalling the main definitions following Kapranov's approach to NC-geometry.

An NC-complete algebra R over a field k , is a k -algebra that is complete with respect to the decreasing "*commutator lie algebra filtration*" on R . The d^{th} filtration $\mathbf{F}^d R$ is generated by expressions of the form

$$a_1 \mathbf{n}_{i_1} a_2 \dots a_n \mathbf{n}_{i_n} a_{n+1}$$

where each $\mathbf{n}_{i_k}$ is formed of i_k nested commutators and such that $i_1 + i_2 + \dots + i_n - n = d$. For example, $\mathbf{F}^0 R = R$ and $\mathbf{F}^1 R$ is generated (*additively*) by elements of the form $a[x_1, [x_2, x_3]]b$ and $a[x_1, x_2]b[x_3, x_4]c$.

Kapranov defines the notions of d -**smoothness** and NC -**smoothness** following the classical definitions of "*formal smoothness*" in algebraic geometry and commutative algebra. In the following, denote by $\mathcal{N}_d$ the category of k -algebras S with $\mathbf{F}^{d+1} S = 0$.

Definition 1.0.1. Let R be a k -algebra, then we say that

- (a) $R \in \mathcal{N}_d$ is d -**smooth** if it is finitely generated and the representable functor $h_{\mathcal{N}_d}^R$ is formally smooth.
- (b) R is NC -**complete** if the canonical map $R \rightarrow \varprojlim R/\mathbf{F}^k R$ is an isomorphism.
- (c) R is said to be NC -**smooth** if it is NC -complete and if $R/\mathbf{F}^{d+1} R$ is d -smooth for all d .

Next, we would like to define the category of **affine NC-schemes** as certain ringed spaces, where the underlying topological space is the *spectrum* of the abelianization of a given NC -complete algebra. We start by defining the appropriate sheaf corresponding to an NC -nilpotent algebra R .

Lemma 1.0.2. Let R be an NC -nilpotent algebra and $\pi : R \rightarrow R_{ab}$ the quotient onto its abelianization. Then for any multiplicative subset $S \subset R_{ab} - \{0\}$, the set $\pi^{-1}(S)$ satisfies the **Ore** conditions for the calculus of fractions.

Using this fact, we define the stalks of the desired ringed space as follows:

Definition 1.0.3. Let R be an NC -nilpotent algebra, we define the associated NC -nilpotent scheme $\mathrm{Spf}(R) := (\mathrm{spec}(R_{ab}), \tilde{\mathcal{O}})$ with the sheaf rings $\tilde{\mathcal{O}}$ defined as the sheaf associated to the presheaf defined on the basis of standard open subsets via

$$\Gamma(D(g), \tilde{\mathcal{O}}) := R[g^{-1}],$$

where $R[g^{-1}]$ denotes the *noncommutative* localization of R by a lift of the element $g \in R_{ab}$.

Proposition 1.0.4. Given an affine NC -nilpotent scheme X as above, then we have:

- (a) The stalk $\tilde{\mathcal{O}}_{\mathfrak{p}}$ is a local ring.
- (b) $\Gamma(X, \tilde{\mathcal{O}}) = R$.

Now given an NC -complete algebra R , we define $\mathrm{Spf}(R)_d$ to be the affine NC -nilpotent scheme associated to $R/\mathbf{F}^{d+1} R$. The NC -complete affine scheme associated to R is now defined as follows:

Definition 1.0.5. Given an NC -complete algebra R , the associated **affine NC-scheme** has underlying topological space $\mathrm{Spf}(R) := (\mathrm{Spec}(R_{ab}), \tilde{\mathcal{O}})$ where

$$\tilde{\mathcal{O}} := \varprojlim \mathcal{O}_{\mathrm{Spf}(R)_d}.$$

An NC -scheme is a locally ringed space that is locally isomorphic to an **affine NC-scheme**.